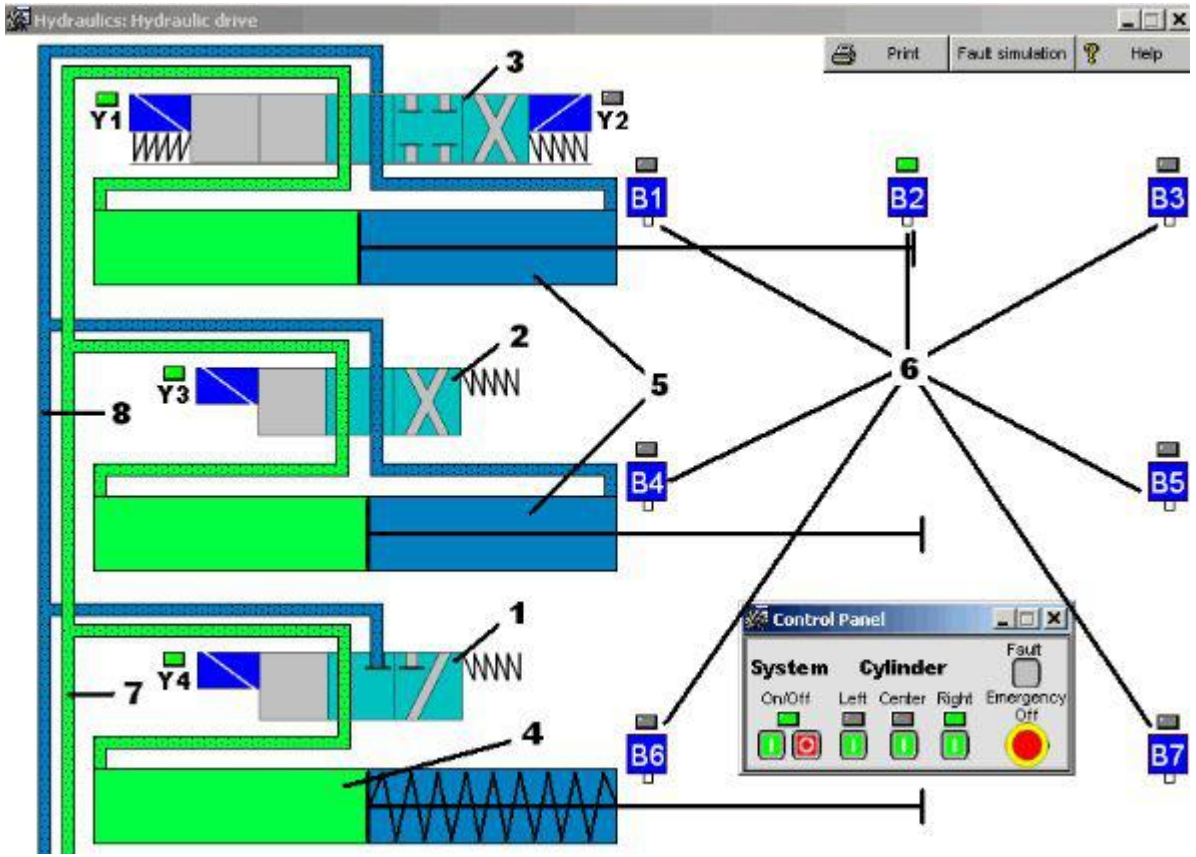
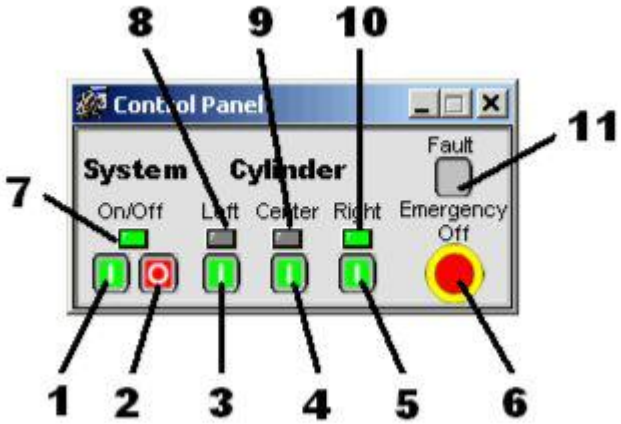
UPRAVLJAČKI ALGORITMI U REALNOM VREMENU
HIDRAULIČNI POGON

Elementi sistema



1. Elektromagnetni 3/2 putni ventil sa povratnom oprugom
2. Elektromagnetni 4/2 putni ventil sa povratnom oprugom
3. Elektromagnetni 4/5 putni ventil sa povratnom oprugom
4. Jednostrano delujući cilindar sa povratnom oprugom
5. Dvostrano delujući cilindar
6. Položajno osetljivi prekidači (granični ili induktivni senzori)
7. Cevovod za pritisno ulje od hidraulične pumpe
8. Povratna linija do rezervoara za ulje

Opis rada sistema

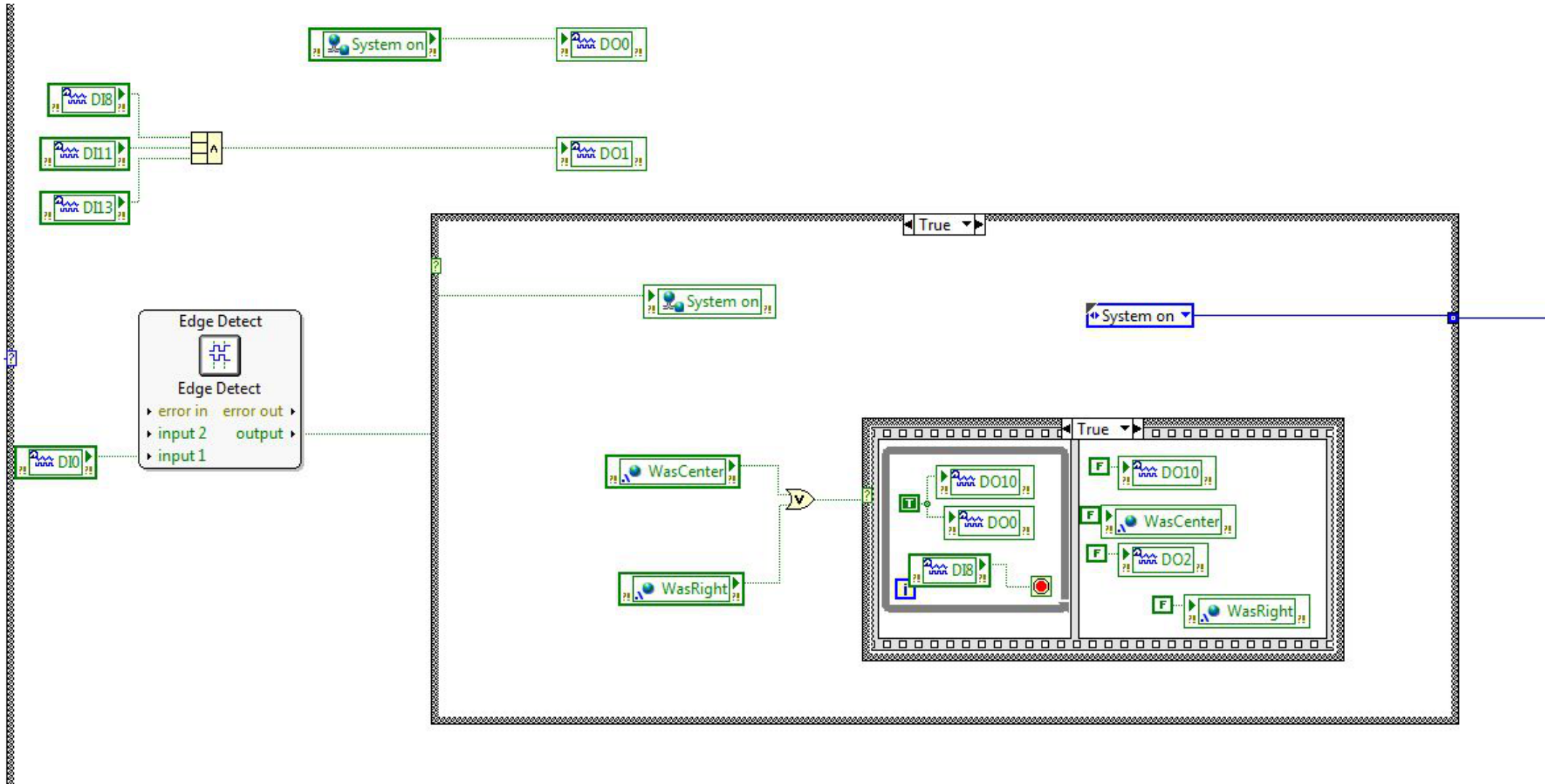


- Sistem se uključuje pritiskom na taster S1. Ovo stanje je označeno time što je LED dioda H1 neprekidno uključena.
- Pokretanje cilindra moguće je pomoću jednog od tri tastera — Levi, Srednji i Desni — čime se cilindar pomera u odgovarajući položaj.
- Kada cilindar dostigne željeni položaj, to se signalizira pomoću LED dioda H2 – H4.
- U slučaju da se pritisne taster Isključivanje (S2), aktuator treba da se isključi nakon završetka tekućeg radnog hoda.
- Ako se pritisne taster za hitno isključenje (Emergency-Off), svi aktuatori moraju biti odmah isključeni.
- U kontrolnom programu mora biti uključena rutina za nadzor krajnjih položaja, kojom se proverava da li cilindar, nakon aktiviranja odgovarajućeg elektromagnetnog ventila, dosegne relevantan krajnji položaj u predviđenom vremenskom intervalu (detektovano pomoću senzora B1 – B7).

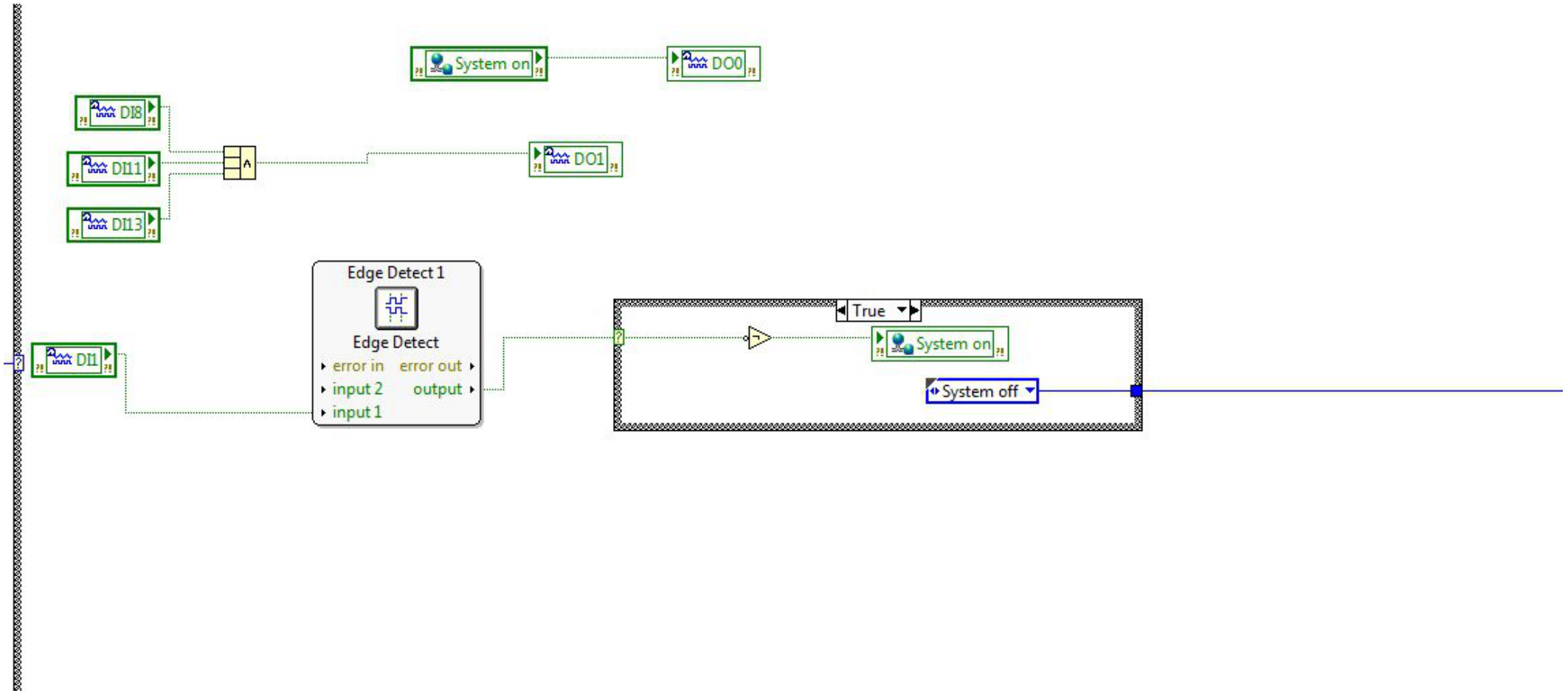
Ulazi i izlazi

Ulaz	Signal	Izlaz	Signal
DI0	S1-System On	DO0	H1- System on
DI1	S2-System off	DO1	H2-Cyl.retracted
DI2	S3-Retract cylinder	DO2	H3-Cyl.interm.pos.
DI3	S4-Cyl.interm.pos	DO3	H4-Cyl.extended
DI4	S5-Extend cylinder	DO4	H5-Fault
DI5	S6-Emergency off	DO9	Y1-Extend cyl.1
DI8	B1-Cyl.1,back	DO10	Y2-Retract cyl.1
DI9	B2-Cyl.1,center	DO11	Y3-Extend cyl.2
DI10	B3-Cyl.1,forward	DO12	Y4-Extend cyl.3
DI11	B4-Cyl.2,back		
DI12	B5-Cyl.2 ,forward		
DI13	B6-Cyl.3, back		
DI14	B7-Cyl.3,forward		

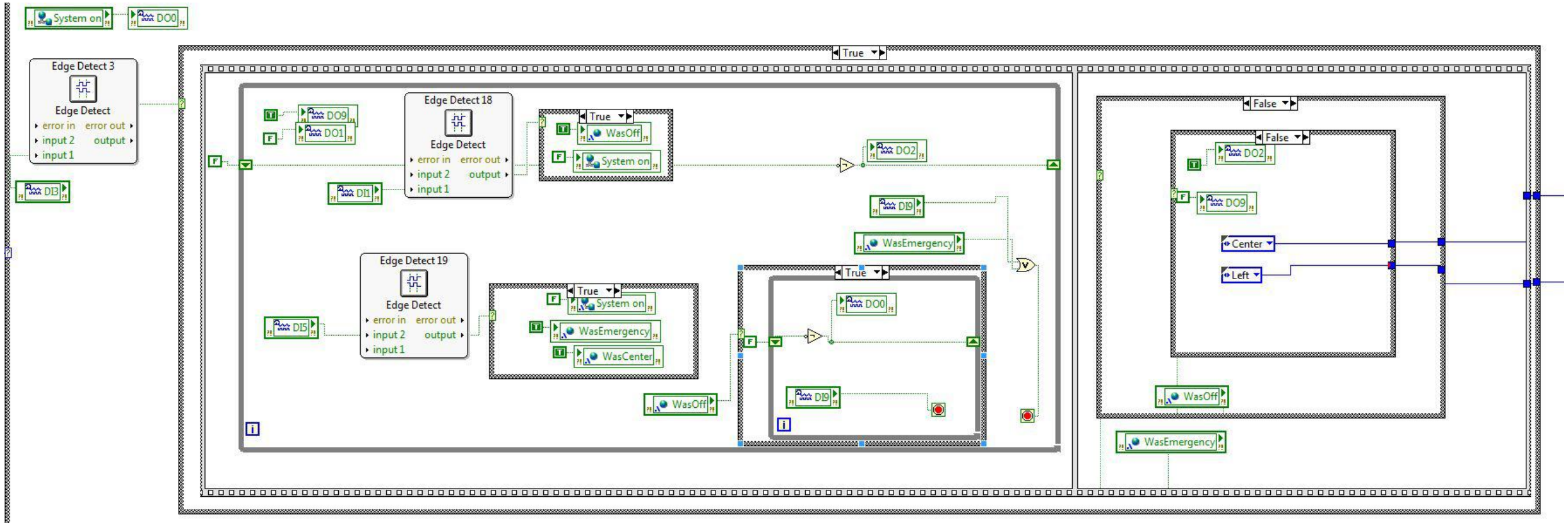
Stanje: System off



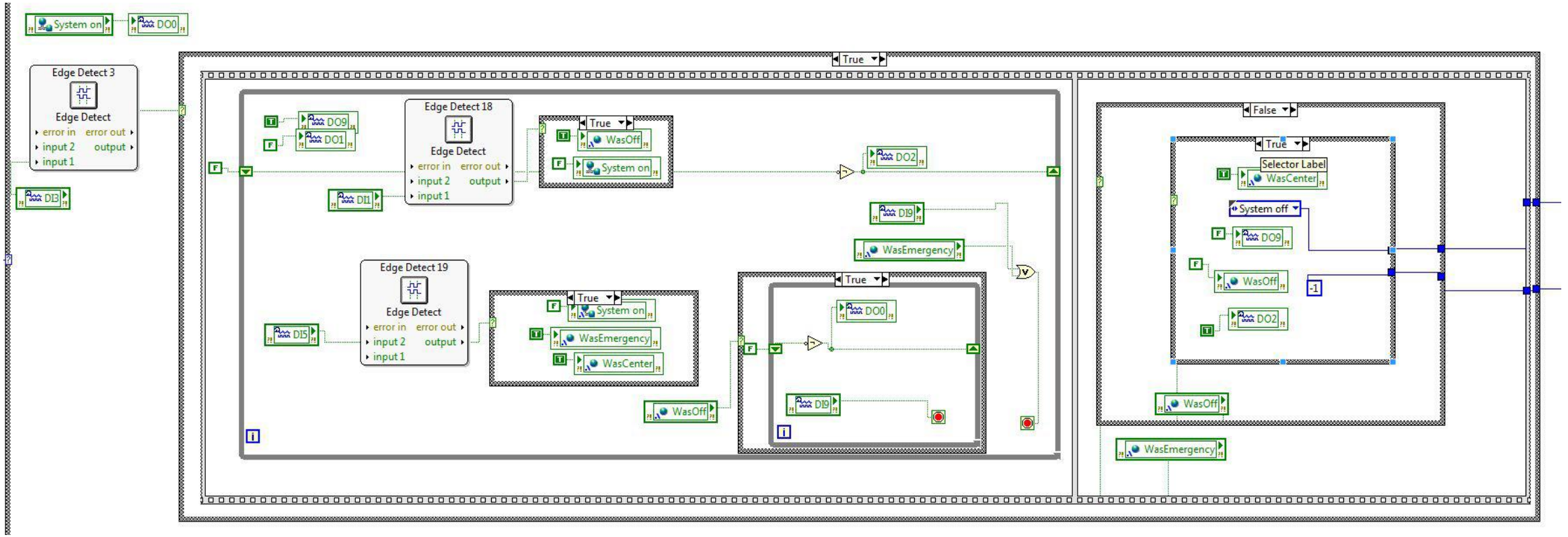
Stanje: System on



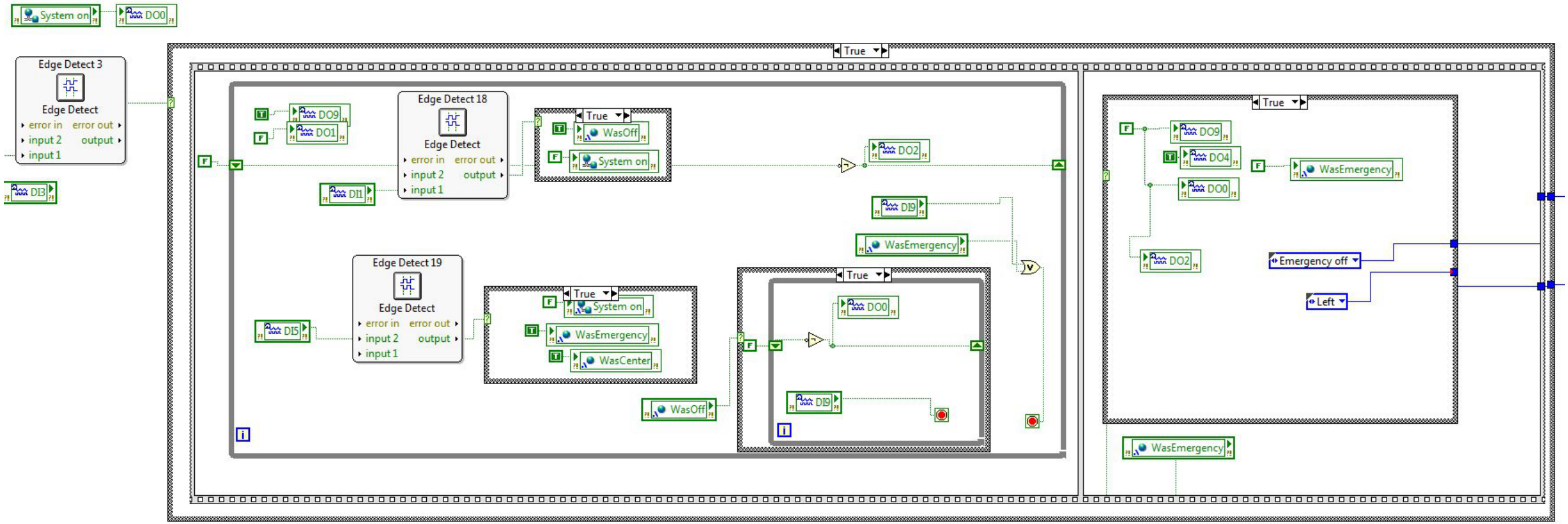
Stanje: Left-Centar



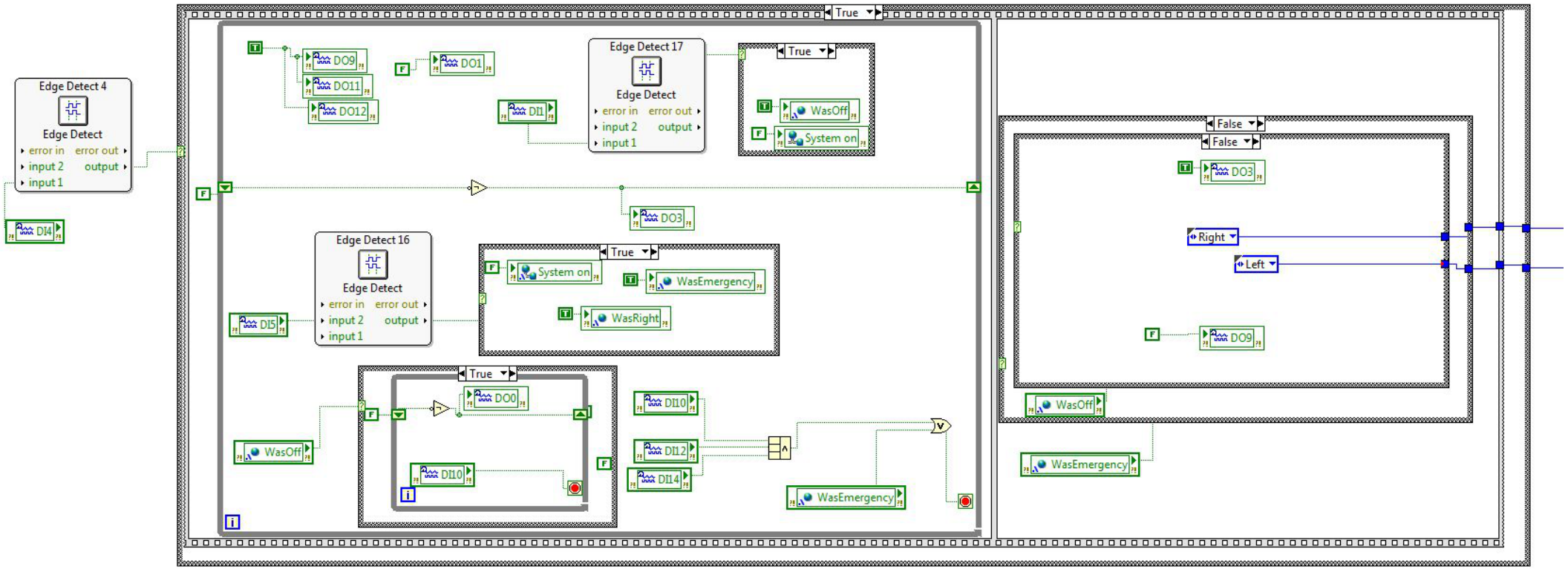
Stanje: Left-Centar



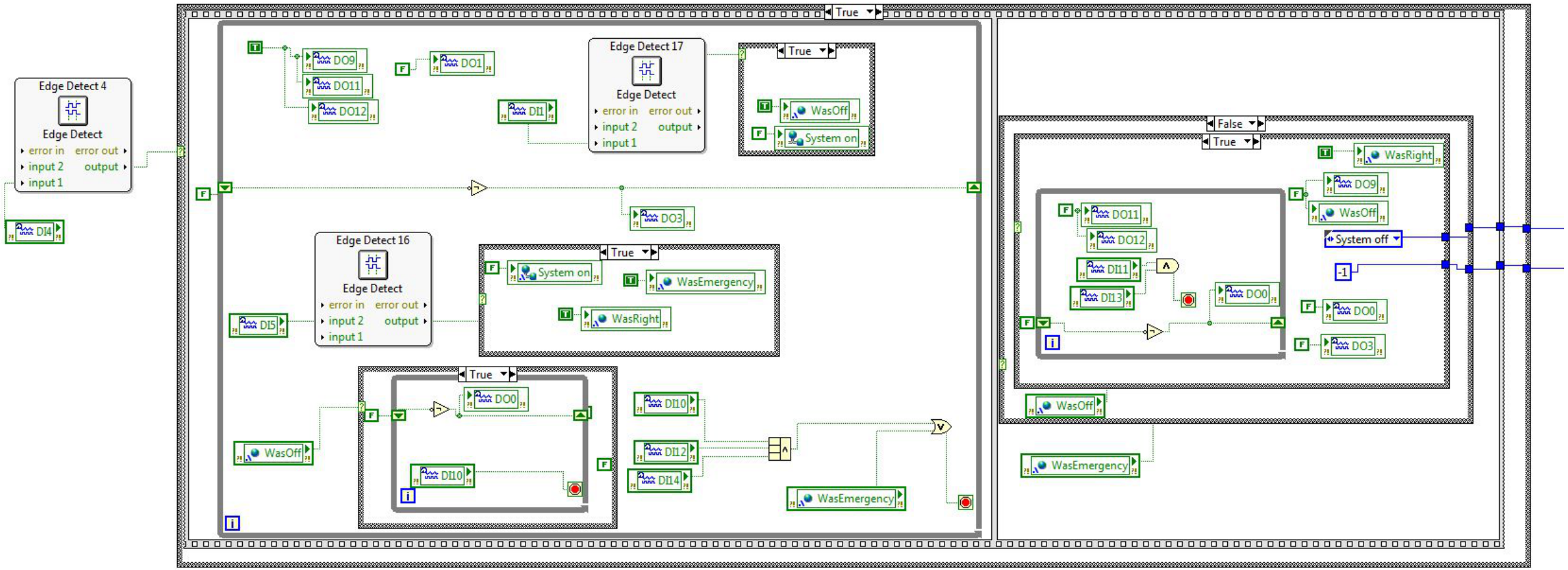
Stanje: Left-Centar



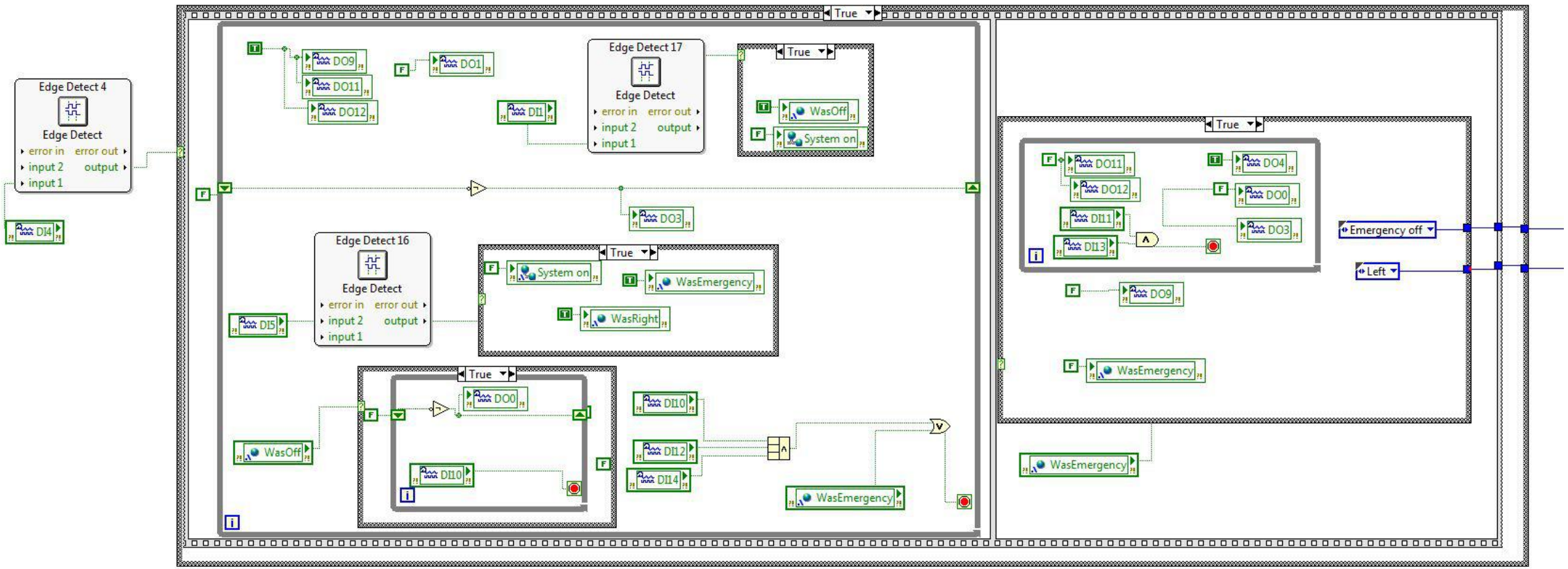
Stanje: Left-Right



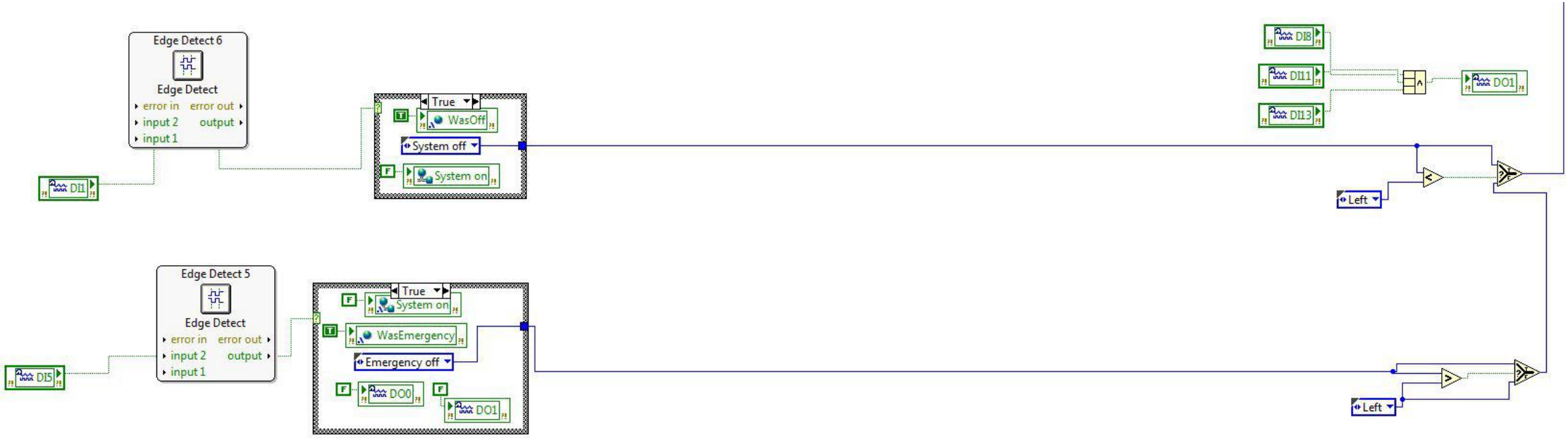
Stanje: Left-Right



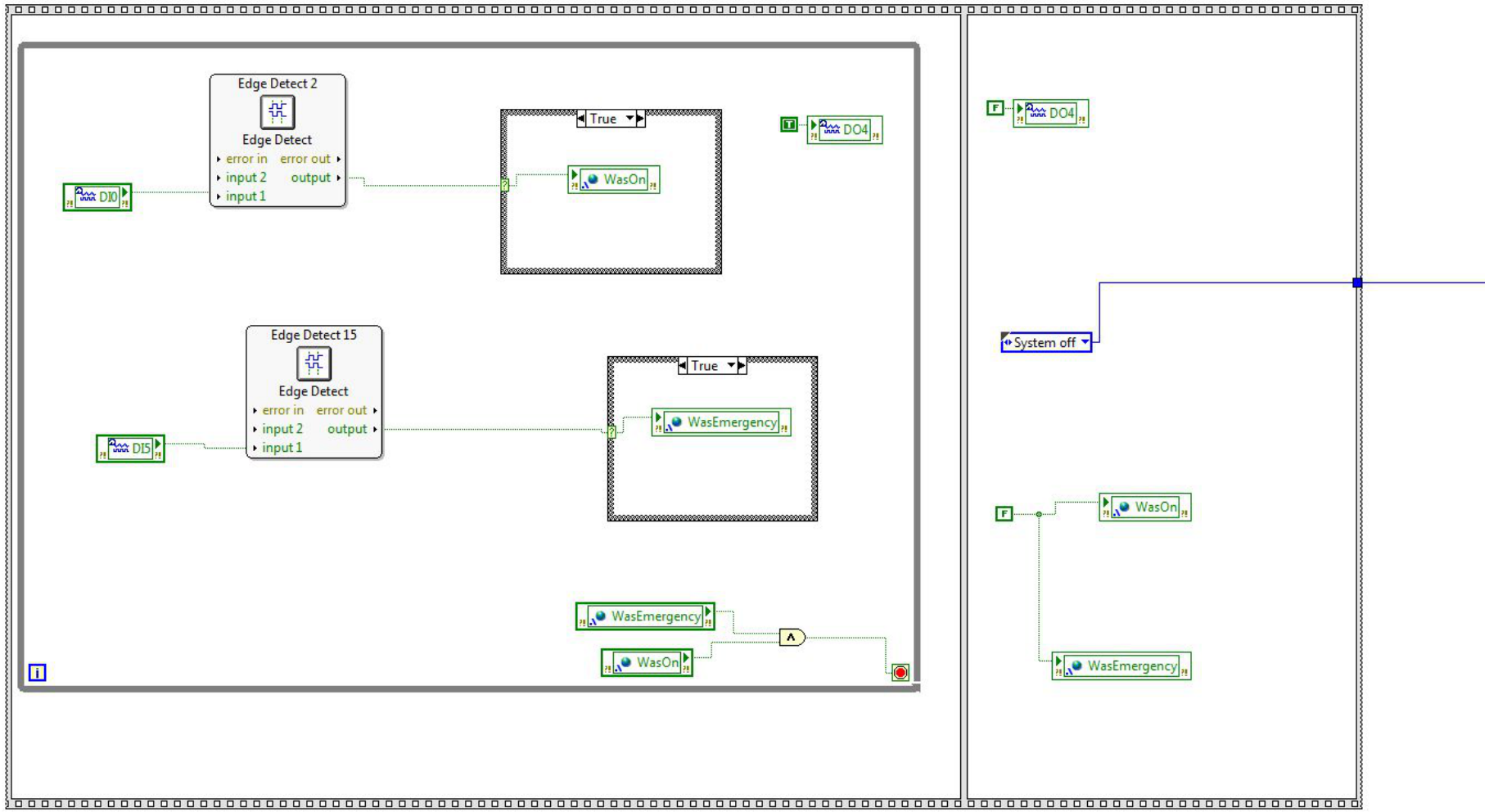
Stanje: Left-Right



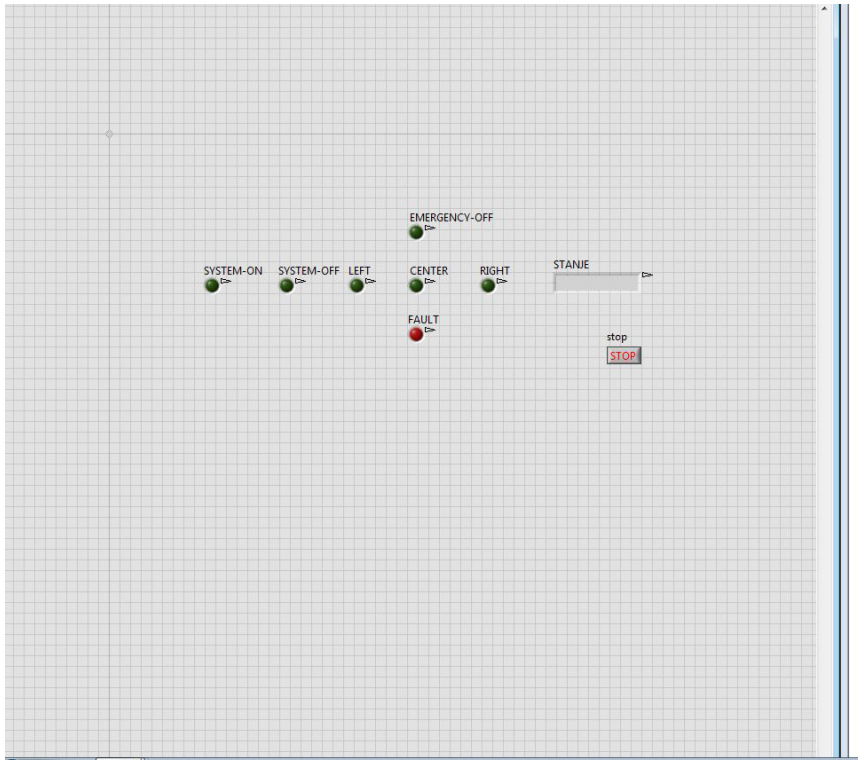
Left-off i Left-Emergency off:

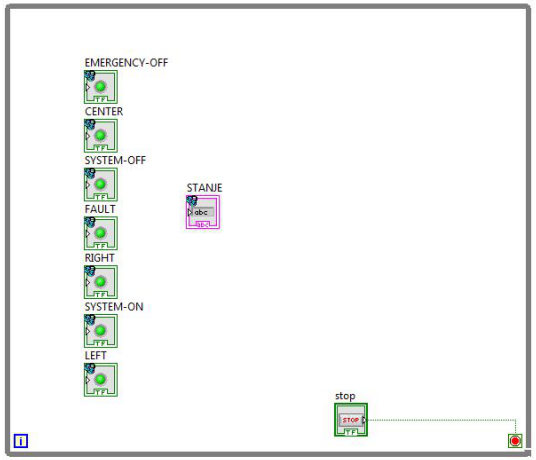


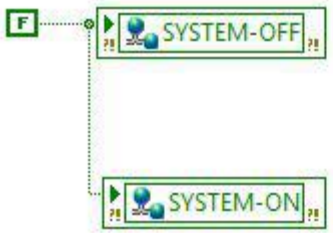
Stanje: Emergency off

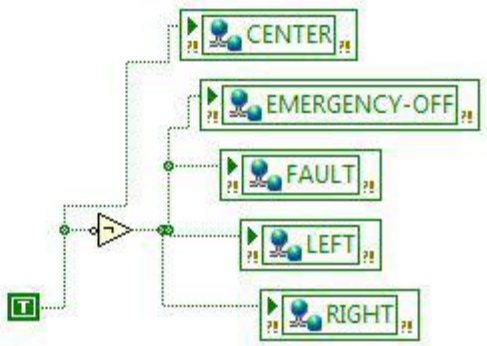


SCADA.VI:


EMERGENCY-OFF
SYSTEM-ON
SYSTEM-OFF
LEFT
CENTER
RIGHT
STANJE
FAULT
stop
STOP


EMERGENCY-OFF
CENTER
SYSTEM-OFF
STANJE
FAULT
RIGHT
SYSTEM-ON
LEFT
stop
STOP


F
SYSTEM-OFF
SYSTEM-ON


CENTER
EMERGENCY-OFF
FAULT
LEFT
RIGHT
T

HVALA NA PAŽNJI!

